

Unit 2 - Linear Algebra Matrices

MSDS 520

Meharry Medical College

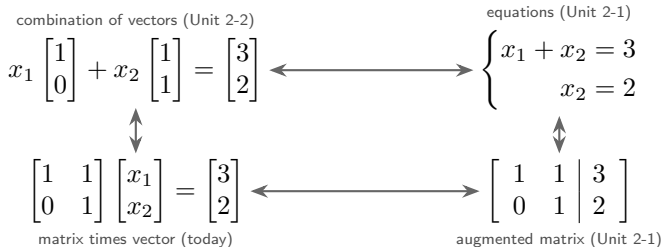
Outline

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- 4 Matrix Arithmetic
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Where Vectors Left Off

A Dependence Problem, Four Ways

Unit 2-2 ended here: is $\bar{u} = \begin{bmatrix} 3 \\ 2 \end{bmatrix}$ a **combination** of $\bar{v} = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$ and $\bar{w} = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$? If so, $\bar{u}$ is redundant.



Naming the pieces: $A\bar{x} = \bar{b}$

$$A = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}$$

The **coefficient matrix**. Its columns are $\bar{v}$ and $\bar{w}$.

$$\bar{x} = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$$

The **unknowns**: how much of each column.

$$\bar{b} = \begin{bmatrix} 3 \\ 2 \end{bmatrix}$$

The **target**: the right-hand side, here $\bar{u}$.

$A\bar{x} = \bar{b}$ reads: **weight the columns of A by $\bar{x}$, and hit $\bar{b}$.**

Why Matrices?

A Problem to Motivate Us

A matinee, from Unit 2-1

"A theater sold 300 tickets: children's at \$6, adults' at \$10, seniors' at \$8, and took in \$2540. As many adult tickets were sold as children's and seniors' combined. How many of each?"

Unit 2-1: three equations.

Today: **one** equation, $A\bar{x} = \bar{b}$.

$$\begin{cases} c + a + s = 300 \\ 6c + 10a + 8s = 2540 \\ -c + a - s = 0 \end{cases}$$

$$\underbrace{\begin{bmatrix} 1 & 1 & 1 \\ 6 & 10 & 8 \\ -1 & 1 & -1 \end{bmatrix}}_A \underbrace{\begin{bmatrix} c \\ a \\ s \end{bmatrix}}_{\bar{x}} = \underbrace{\begin{bmatrix} 300 \\ 2540 \\ 0 \end{bmatrix}}_{\bar{b}}$$

Where we are headed

- A holds the **prices and rules**, $\bar{x}$ the unknown **counts**, $\bar{b}$ the **totals**.
- By the end of **Matrix Times Vector**, one line of code solves it.

Where This Shows Up in Data Science

Almost everything you compute with is a matrix

- **A data table is a matrix.** One row per record, one column per feature. Unit 2-2 read the rows as vectors; now the whole table is one object.
- **A network is a matrix.** Facebook friendships, web links, and gene interactions are all stored as a grid of who connects to whom.
- **Fitting a model is a matrix product.** Least-squares regression is built from $X^T X$, a product you will be able to compute by the end of today.
- **Moving an image is matrix multiplication.** Classifiers train on rotated and flipped copies of every picture (**augmentation**), all made this way.
- **PCA**, this unit's capstone, starts from a **symmetric** covariance matrix and **rotates** the axes to fit the data.
- **A neural network layer** is a matrix transformation followed by a simple nonlinear step.

Unit 2-1 already used a grid of numbers, the augmented matrix. Today the grid becomes an object with its own arithmetic.

What We Will Build

This deck: the machinery

Matrices, their arithmetic, and their **products**.

- Mostly **technical**: definitions, notation, and the rules of matrix arithmetic.
- Little application or geometry, **on purpose**: systems, vectors, dot products, and sums all have to be joined into one tool first.
- Expect to **compute** more than to picture.

Unit 2-4: putting it to work

A matrix as a **function** that moves the plane.

- Much more **geometric**: a matrix moves the whole plane, and you can watch it happen.
- Much more **applicable**: rotations, projections, and the road to PCA.
- Much more **intuitive**: the rules from this deck finally get pictures.

If this deck feels dry, that is expected: Unit 2-4 puts the tool to work.

Notebooks: U2-3_Matrices-1, -2.

Matrices and Their Entries

What Is a Matrix?

Definition

An $m \times n$ **matrix** is a rectangular grid of numbers with m **rows** and n **columns**.

$$A = \underbrace{\begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{bmatrix}}_{n = 3 \text{ columns}} \left. \vphantom{\begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{bmatrix}} \right\} \begin{matrix} m = 2 \\ \text{rows} \end{matrix}$$

A 2×3 matrix.

$$\begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{bmatrix} \quad \begin{matrix} \text{row 1} \\ \text{row 2} \end{matrix}$$

column 1

A **row** runs across; a **column** runs down.

Size is always rows $\times$ columns

- Rows first, always: this A is 2×3 , never 3×2 .
- A 3×1 matrix is **tall**; a 1×3 matrix is **wide**.
- When $m = n$ the matrix is **square**.

Naming Entries

Every entry has an address:

$$A = \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \end{bmatrix}$$

a_{ij} sits in **row** i , **column** j . The first subscript is the row, the second the column.

$$A = \begin{bmatrix} 1 & \textcircled{2} & 3 \\ 4 & 5 & \textcircled{6} \end{bmatrix}$$

$a_{12} = 2$
 $a_{23} = 6$

Row 1, column 2; row 2, column 3.

The naming convention: the letter tells you what it is

- **Capital letter: a matrix.** A , B , M .
- **Same letter, lowercase, two subscripts: its entries.** a_{ij} lives in A , b_{ij} in B .
- **Lowercase with a bar: a vector.** $\bar{v}$, $\bar{x}$, as in Unit 2-2.
- **Plain lowercase: a number (a scalar).** c , k .

A Vector Is a Matrix Too

The column vectors of Unit 2-2 are matrices with **one column**. A vector can also be written as a single row: a matrix with **one row**.

$$\bar{v} = \left[\begin{array}{c} 3 \\ 4 \\ 1 \end{array} \right] \left. \vphantom{\begin{array}{c} 3 \\ 4 \\ 1 \end{array}} \right\} \begin{array}{l} 3 \text{ rows} \\ \\ 1 \text{ column} \end{array}$$

Column vector: 3×1 , **tall**.

$$\bar{w} = \left[\begin{array}{ccc} 2 & 7 & 5 \end{array} \right] \left. \vphantom{\begin{array}{ccc} 2 & 7 & 5 \end{array}} \right\} \begin{array}{l} 1 \text{ row} \\ \\ 3 \text{ columns} \end{array}$$

Row vector: 1×3 , **wide**.

Two shapes of vector

- The **column** is the default: unless we say otherwise a vector is a column, so $\bar{v} \in \mathbb{R}^n$ is $n \times 1$.
- A **row** vector such as $\bar{w}$ is $1 \times n$. Rows meet columns when we multiply, later today.
- Every rule for matrices today applies to vectors too. Nothing new to learn.

When Are Two Matrices Equal?

Equal means: same size, and every matching entry agrees

For vectors and matrices alike: check the **size** first, then $a_{ij} = b_{ij}$ for all i, j .

	Vectors	Matrices
Equal	$\begin{bmatrix} 3 \\ 4 \end{bmatrix} = \begin{bmatrix} 3 \\ 4 \end{bmatrix}$	$\begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix} = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$
Not equal: same numbers, wrong places	$\begin{bmatrix} 3 \\ 4 \end{bmatrix} \neq \begin{bmatrix} 4 \\ 3 \end{bmatrix}$	$\begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix} \neq \begin{bmatrix} 1 & 3 \\ 2 & 4 \end{bmatrix}$
Not equal: sizes differ, so stop there	$\begin{bmatrix} 3 \\ 4 \end{bmatrix} \neq \begin{bmatrix} 3 & 4 \end{bmatrix}$	$\begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{bmatrix} \neq \begin{bmatrix} 1 & 4 \\ 2 & 5 \\ 3 & 6 \end{bmatrix}$

A row is never a column, whatever the numbers. And one matrix equation is one equation per entry: $\begin{bmatrix} x & 2 \\ 3 & y \end{bmatrix} = \begin{bmatrix} 5 & 2 \\ 3 & -1 \end{bmatrix}$ means $x = 5$, $y = -1$.

Matrices in Python

A matrix is a 2-D NumPy array

```
import numpy as np
A = np.array([[1, 2, 3],
              [4, 5, 6]])
A.shape      # (2, 3): rows, columns
A[0, 1]      # 2, which is a12
A[1, 2]      # 6, which is a23
A[0, :]      # row 1:    [1 2 3]
A[:, 1]      # column 2: [2 5]
```

Python counts from 0

Mathematics counts from 1,
Python from 0:

$$a_{ij} \text{ is } A[i-1, j-1]$$

This off-by-one is the most
common bug in a first linear
algebra notebook.

Vectors: a 1-D array has no shape until you give it one

```
v = np.array([3, 4, 1])    # shape (3,):   no row or column
v.reshape(3, 1)            # shape (3, 1): a column, 3 x 1
v.reshape(1, 3)            # shape (1, 3): a row,    1 x 3
```

Notebook U2-3_Matrices-1_ArithmeticAndSpecial, §1, entries and dimensions

A Data Table Is a Matrix

Three patients, three measurements (the first two are Unit 2-2's $\bar{p}$ and $\bar{q}$):

	Age	BP	A1c
Patient 1	54	130	6.1
Patient 2	58	138	6.4
Patient 3	41	118	5.6

$\xrightarrow{\text{numbers only}}$

$$X = \begin{bmatrix} 54 & 130 & 6.1 \\ 58 & 138 & 6.4 \\ 41 & 118 & 5.6 \end{bmatrix}$$

The same grid, two views

- Each **row** is one patient: row 1 is patient 1's measurements.
- Each **column** is one feature across everyone: age is a vector in $\mathbb{R}^3$.
- x_{ij} is feature j of patient i : $x_{23} = 6.4$ is patient 2's A1c.

Both views matter

- Similar **patients**: compare rows. Redundant **features**: compare columns.
- One sample per row is how data is **stored**; a lone vector is still a column.

Exercise: Read the Matrix

Try it

Let

$$M = \begin{bmatrix} 3 & 0 & -1 \\ 2 & 7 & 4 \end{bmatrix}, \quad S = \begin{bmatrix} 2 & 5 \\ 5 & -1 \end{bmatrix}.$$

- ① State the dimensions of M , and read off m_{13} and m_{21} .
- ② Can M ever equal S ? Why or why not?
- ③ A patient record with 12 lab values is stored for 250 patients. What are the dimensions of the data matrix, and what does the entry in row 17, column 3 mean?

Exercise: Solution

Answers

- ① M is 2×3 : two rows, three columns. m_{13} is row 1, column 3, which is -1 ; m_{21} is row 2, column 1, which is 2. (Not the other way round: rows first.)
- ② **Never.** M is 2×3 and S is 2×2 . Equality requires the same shape before a single entry is compared.
- ③ 250×12 : one row per patient, one column per lab value. Row 17, column 3 is the **third lab value of the seventeenth patient**. The row is that patient's feature vector; the column is that lab value across all 250 patients.

Matrix Arithmetic

Addition and Scalar Multiplication

Entry by entry, exactly as for vectors

Matrices of the **same shape** add entry by entry, and a scalar multiplies every entry:

$$(A + B)_{ij} = a_{ij} + b_{ij}, \quad (cA)_{ij} = c a_{ij}.$$

Subtraction is $A - B = A + (-1)B$; vectors are the one-column case.

$$\begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix} + \begin{bmatrix} 5 & 6 \\ 7 & 8 \end{bmatrix} = \begin{bmatrix} 6 & 8 \\ 10 & 12 \end{bmatrix}, \quad 2 \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix} = \begin{bmatrix} 2 & 4 \\ 6 & 8 \end{bmatrix}$$

Rules from numbers

- $A + B = B + A$
- $(A + B) + C = A + (B + C)$
- $c(A + B) = cA + cB$, and $A + 0 = A$

Watch out: the size rule from equality

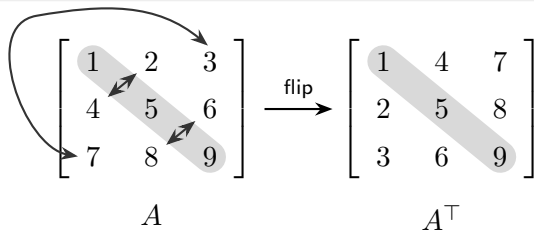
Adding and subtracting need the **same size**, just like equality. A 2×2 plus a 2×3 is **undefined**, and so is a column plus a row, $\begin{bmatrix} 3 \\ 4 \end{bmatrix} + \begin{bmatrix} 3 & 4 \end{bmatrix}$. numpy errors on the first but silently **broadcasts** the second to a 2×2 grid.

Notebook U2-3_Matrices-1_ArithmeticAndSpecial, §2, sums and scalar multiples

The Transpose

Definition: transpose

The **transpose** A^T swaps rows and columns: $(A^T)_{ij} = a_{ji}$. The grid flips over its **main diagonal** $(a_{11}, a_{22}, \dots)$, so an $m \times n$ matrix becomes $n \times m$.



Not square? The shape flips too:

$$\begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{bmatrix}^T = \begin{bmatrix} 1 & 4 \\ 2 & 5 \\ 3 & 6 \end{bmatrix}$$

The diagonal (shaded) stays put; every other entry swaps with its mirror image.

Rules

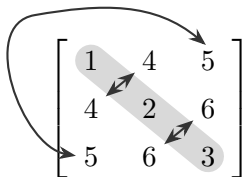
$$(A^T)^T = A, \quad (A + B)^T = A^T + B^T, \quad (cA)^T = cA^T.$$

Notebook U2-3_Matrices-1_ArithmeticAndSpecial, §3, the transpose

Symmetric Matrices

Definition: symmetric

A square matrix A is **symmetric** when $A^T = A$, that is, $a_{ij} = a_{ji}$ for every i and j : a **mirror image** across the diagonal, so the flip changes nothing.



$$S = S^T$$

Every pair that trades places holds the same number.

Worth knowing

- Only a **square** matrix can be symmetric: the flip has to keep the size.
- **Not** symmetric: $\begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$, since $a_{12} = 2$ but $a_{21} = 3$.
- The **covariance matrix** behind PCA is symmetric, and so is a **friendship** matrix (later today).

Exercise: Arithmetic and Transposes

Try it

Let $A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$, $B = \begin{bmatrix} 0 & -1 \\ 2 & 5 \end{bmatrix}$, $C = \begin{bmatrix} 1 & 0 & 2 \\ 3 & 1 & 0 \end{bmatrix}$.

- ① Compute $A + B$ and $3A - B$.
- ② Write down A^T and C^T , with their dimensions.
- ③ Compute $A + A^T$. Is it symmetric? Would that be true for **any** square A ?
- ④ What is wrong with $A + C$?

Exercise: Solution

Answers

① $A + B = \begin{bmatrix} 1 & 1 \\ 5 & 9 \end{bmatrix}$. $3A = \begin{bmatrix} 3 & 6 \\ 9 & 12 \end{bmatrix}$, so $3A - B = \begin{bmatrix} 3 & 7 \\ 7 & 7 \end{bmatrix}$. Scale first, then subtract entry by entry.

② $A^T = \begin{bmatrix} 1 & 3 \\ 2 & 4 \end{bmatrix}$, still 2×2 . $C^T = \begin{bmatrix} 1 & 3 \\ 0 & 1 \\ 2 & 0 \end{bmatrix}$, now 3×2 : the rows of C have become its columns.

③ $A + A^T = \begin{bmatrix} 2 & 5 \\ 5 & 8 \end{bmatrix}$, which **is** symmetric. And it always is:
 $(A + A^T)^T = A^T + (A^T)^T = A^T + A$, the same matrix. Adding a square matrix to its own transpose is the standard way to **manufacture** a symmetric matrix.

④ **The shapes differ.** A is 2×2 and C is 2×3 ; there is no third column of A to add to the third column of C . The sum is undefined.

Block Notation

Block Notation: a Matrix by Its Columns or Its Rows

The same matrix can be grouped into vectors in two ways:

By columns

$$\begin{bmatrix} \boxed{1} & \boxed{2} \\ \boxed{3} & \boxed{4} \end{bmatrix}$$

$\bar{c}_1$ $\bar{c}_2$

$$A = [\bar{c}_1 \quad \bar{c}_2], \quad \bar{c}_1 = \begin{bmatrix} 1 \\ 3 \end{bmatrix}, \quad \bar{c}_2 = \begin{bmatrix} 2 \\ 4 \end{bmatrix}$$

By rows

$$\begin{bmatrix} \boxed{1} & \boxed{2} \\ \boxed{3} & \boxed{4} \end{bmatrix} \begin{matrix} \bar{r}_1^\top \\ \bar{r}_2^\top \end{matrix}$$

$$A = \begin{bmatrix} \bar{r}_1^\top \\ \bar{r}_2^\top \end{bmatrix}, \quad \bar{r}_1 = \begin{bmatrix} 1 \\ 2 \end{bmatrix}, \quad \bar{r}_2 = \begin{bmatrix} 3 \\ 4 \end{bmatrix}$$

What the notation means

- The brackets now hold **vectors**, not numbers: this is **block notation**.
- A vector is always a **column**, so row i is written as a column **transposed**: $\bar{r}_i^\top, 1 \times n$.
- Any size works: an $m \times n$ matrix is n columns side by side, $[\bar{c}_1 \quad \bar{c}_2 \quad \cdots \quad \bar{c}_n]$, or m rows stacked.

Block Notation: Blocks That Are Matrices

Blocks can be matrices too. Cut a 4×4 matrix down the middle, both ways:

$$M = \begin{bmatrix} \overset{M_{11}}{\boxed{\begin{matrix} 1 & 2 \\ 5 & 6 \end{matrix}}} & \overset{M_{12}}{\boxed{\begin{matrix} 3 & 4 \\ 7 & 8 \end{matrix}}} \\ \underset{M_{21}}{\boxed{\begin{matrix} 9 & 10 \\ 13 & 14 \end{matrix}}} & \underset{M_{22}}{\boxed{\begin{matrix} 11 & 12 \\ 15 & 16 \end{matrix}}} \end{bmatrix}$$

Four 2×2 blocks

$$M = \begin{bmatrix} M_{11} & M_{12} \\ M_{21} & M_{22} \end{bmatrix}$$

$$M_{11} = \begin{bmatrix} 1 & 2 \\ 5 & 6 \end{bmatrix}, \quad M_{12} = \begin{bmatrix} 3 & 4 \\ 7 & 8 \end{bmatrix}$$

$$M_{21} = \begin{bmatrix} 9 & 10 \\ 13 & 14 \end{bmatrix}, \quad M_{22} = \begin{bmatrix} 11 & 12 \\ 15 & 16 \end{bmatrix}$$

The rules for blocks

- M_{12} is the block in block-row 1, block-column 2: **rows first**, as for entries.
- Each cut runs **all the way across**, so blocks side by side share a height and blocks stacked share a width.
- Columns and rows are the special case: blocks of size $m \times 1$ or $1 \times n$.

Matrix Times Vector

Where the Product Comes From

The question (Unit 2-2)

What weights c_1, c_2 reach $\bar{p} = \begin{bmatrix} 3 \\ 3 \end{bmatrix}$? As a combination, or as a system:

$$c_1 \begin{bmatrix} 2 \\ 1 \end{bmatrix} + c_2 \begin{bmatrix} -1 \\ 1 \end{bmatrix} = \begin{bmatrix} 2c_1 - c_2 \\ c_1 + c_2 \end{bmatrix} = \begin{bmatrix} 3 \\ 3 \end{bmatrix} \iff \begin{cases} 2c_1 - c_2 = 3 \\ c_1 + c_2 = 3 \end{cases}$$

The fix: give the pieces names

Both forms scatter the unknowns. Instead, the basis vectors become the **columns** of B , the weights stack into $\bar{c}$, and $B\bar{c}$ is **defined** as that combination:

$$\underbrace{\begin{bmatrix} 2 & -1 \\ 1 & 1 \end{bmatrix}}_B \underbrace{\begin{bmatrix} c_1 \\ c_2 \end{bmatrix}}_{\bar{c}} = c_1 \begin{bmatrix} 2 \\ 1 \end{bmatrix} + c_2 \begin{bmatrix} -1 \\ 1 \end{bmatrix}$$

Not Multiplication as You Know It

In **arithmetic**, multiplying takes two numbers and gives a number (the **same kind** in and out), and **order does not matter**: $3 \times 4 = 4 \times 3$.

Linear algebra's "products" break those habits

	In	Out	Swap the order?
3×4	two numbers	a number	same
$A\bar{x}$	$m \times n, n \times 1$	$m \times 1$	undefined
$\bar{v} \cdot \bar{w}$	two vectors	a number	same
AB (soon)	two $n \times n$	$n \times n$	usually different

Watch out

- A vector **is** a matrix ($n \times 1$), so A and $\bar{x}$ are the same kind of thing, but their **shapes** must fit.
- "Product" is a name **by analogy**: built from ordinary $\times$ and $+$, but arithmetic's rules do **not** come along.

Matrix Times Vector: the Definition

Definition

A ($m \times n$) times $\bar{x}$ ($n \times 1$) is the $m \times 1$ vector with entries $(A\bar{x})_i = \sum_{j=1}^n a_{ij} x_j$.

For 2×2 :

$$\begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} a_{11}x_1 + a_{12}x_2 \\ a_{21}x_1 + a_{22}x_2 \end{bmatrix}$$

Example:

$$A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}, \bar{x} = \begin{bmatrix} 5 \\ 6 \end{bmatrix}$$

Row view: how you **compute**

Group the terms **across each row**:

$$A\bar{x} = \begin{bmatrix} \bar{r}_1^\top \\ \bar{r}_2^\top \end{bmatrix} \bar{x} = \begin{bmatrix} \bar{r}_1 \cdot \bar{x} \\ \bar{r}_2 \cdot \bar{x} \end{bmatrix}$$
$$\begin{bmatrix} 1 \cdot 5 + 2 \cdot 6 \\ 3 \cdot 5 + 4 \cdot 6 \end{bmatrix} = \begin{bmatrix} 17 \\ 39 \end{bmatrix}$$

Column view: how you **think**

Group the terms **by x_1 and x_2** :

$$A\bar{x} = \begin{bmatrix} \bar{c}_1 & \bar{c}_2 \end{bmatrix} \bar{x} = x_1 \bar{c}_1 + x_2 \bar{c}_2$$
$$5 \begin{bmatrix} 1 \\ 3 \end{bmatrix} + 6 \begin{bmatrix} 2 \\ 4 \end{bmatrix} = \begin{bmatrix} 17 \\ 39 \end{bmatrix}$$

Matrix Times Vector: Worked, and Impossible

Worked: $(2 \times 2)(2 \times 1)$

$$\begin{bmatrix} 2 & 1 \\ 4 & 3 \end{bmatrix} \begin{bmatrix} 3 \\ 2 \end{bmatrix} = \begin{bmatrix} 2 \cdot 3 + 1 \cdot 2 \\ 4 \cdot 3 + 3 \cdot 2 \end{bmatrix} = \begin{bmatrix} 8 \\ 18 \end{bmatrix}$$

Worked: $(2 \times 3)(3 \times 1)$, not square

$$\begin{bmatrix} 1 & 0 & 2 \\ 3 & 1 & 1 \end{bmatrix} \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix} = \begin{bmatrix} 1 + 0 + 6 \\ 3 + 2 + 3 \end{bmatrix} = \begin{bmatrix} 7 \\ 8 \end{bmatrix}$$

Sizes must match, or there is no product

$(m \times n) \underbrace{(n \times 1)}_{\text{equal}} = (m \times 1)$: a row of A must be **exactly as long** as $\bar{x}$.

Impossible:

- $\begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix} \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}$ is $(2 \times \mathbf{2})(\mathbf{3} \times 1)$: each row has 2 entries, $\bar{x}$ has 3. The 3 has nothing to pair with.
- $\begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{bmatrix} \begin{bmatrix} 1 \\ 2 \end{bmatrix}$ is $(2 \times \mathbf{3})(\mathbf{2} \times 1)$: each row has 3 entries, $\bar{x}$ only 2.

Undefined means undefined: no zero-padding, no partial answer. `numpy` raises an error.

Line It Up: Where Each Entry of $A\bar{x}$ Goes

$$A \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix} \begin{array}{c} \text{---} \\ \text{---} \end{array} \begin{bmatrix} 17 \\ 39 \end{bmatrix} A\bar{x}$$

The diagram illustrates the matrix multiplication $A\bar{x}$. Matrix A is a 2x2 matrix with entries 1, 2, 3, and 4. The entry 4 is highlighted with a solid box. Vector $\bar{x}$ is a 2x1 matrix with entries 5 and 6, shown in a dashed box. A dashed line connects the box around 4 in A to the circle around 39 in $A\bar{x}$, indicating that 39 is the result of the dot product of the second row of A and the vector $\bar{x}$.

How to use it

- Put $\bar{x}$ **above** and A to the **left**; the answer $A\bar{x}$ fills the corner.
- Row i of A runs **across**, $\bar{x}$ runs **down**, and they meet at entry i of $A\bar{x}$.
- That entry is their dot product: row 2 gives $3 \cdot 5 + 4 \cdot 6 = 39$.

The layout shows the shape rule too: a row of A must be exactly as long as $\bar{x}$, or the pairs do not line up.

Side by Side Now Means Something

Notation, updated from Unit 2-2

Written	Shapes	Meaning
$A\bar{x}$	$(m \times n)(n \times 1)$	matrix times vector: a vector in $\mathbb{R}^m$
$\bar{x}A$	$(n \times 1)(m \times n)$	undefined : the inner sizes do not match
$\bar{y}^\top A$	$(1 \times m)(m \times n)$	a row , $1 \times n$: $\bar{y}$ dotted with each column
$\bar{v}\bar{w}$	$(n \times 1)(n \times 1)$	still undefined for two columns
$\bar{v}^\top \bar{w}$	$(1 \times n)(n \times 1)$	a number , 1×1 : the dot product $\bar{v} \cdot \bar{w}$
$\bar{v}\bar{w}^\top$	$(n \times 1)(1 \times n)$	an $n \times n$ matrix (outer product, Unit 2-4)

The dot product is a matrix product

A row $\bar{v}^\top$ is a column tipped over, $1 \times n$. Row times column is 1×1 , a number:

$$\bar{v}^\top \bar{w} = \begin{bmatrix} v_1 & v_2 \end{bmatrix} \begin{bmatrix} w_1 \\ w_2 \end{bmatrix} = v_1 w_1 + v_2 w_2 = \bar{v} \cdot \bar{w}, \quad \text{e.g.} \quad \begin{bmatrix} 1 & 2 \end{bmatrix} \begin{bmatrix} 3 \\ 4 \end{bmatrix} = 3 + 8 = 11$$

So each entry of $A\bar{x}$ is a row of A times $\bar{x}$: $(A\bar{x})_i = \bar{r}_i^\top \bar{x}$.

A System Is Now One Equation

Back to the matinee. The augmented matrix of Unit 2-1 was this product in disguise:

$$\begin{cases} c + a + s = 300 \\ 6c + 10a + 8s = 2540 \\ -c + a - s = 0 \end{cases} \longleftrightarrow \underbrace{\begin{bmatrix} 1 & 1 & 1 \\ 6 & 10 & 8 \\ -1 & 1 & -1 \end{bmatrix}}_A \underbrace{\begin{bmatrix} c \\ a \\ s \end{bmatrix}}_{\vec{x}} = \underbrace{\begin{bmatrix} 300 \\ 2540 \\ 0 \end{bmatrix}}_{\vec{b}}$$

Solve it in one line

- `np.linalg.solve(A, b)` gives $c = 80$, $a = 150$, $s = 70$.
- $[A \mid \vec{b}]$ from Unit 2-1 is just A with $\vec{b}$ pinned on.

Check it with the row view

$$A \begin{bmatrix} 80 \\ 150 \\ 70 \end{bmatrix} = \begin{bmatrix} 80 + 150 + 70 \\ 480 + 1500 + 560 \\ -80 + 150 - 70 \end{bmatrix} = \begin{bmatrix} 300 \\ 2540 \\ 0 \end{bmatrix} = \vec{b}$$

Application: A Bill in One Product

Two students take credits at two colleges, each with a fixed price per credit:

Credits taken		
	College 1	College 2
Student 1	10	7
Student 2	11	4

Cost per credit	
	\$ per credit
College 1	60
College 2	80

Credits are a matrix, prices a vector, and their product is every bill

$$\underbrace{\begin{bmatrix} 10 & 7 \\ 11 & 4 \end{bmatrix}}_{\text{credits}} \underbrace{\begin{bmatrix} 60 \\ 80 \end{bmatrix}}_{\text{cost per credit}} = \underbrace{\begin{bmatrix} 1160 \\ 980 \end{bmatrix}}_{\text{bills}}$$

- **Row view:** student 1 pays $10(60) + 7(80) = 1160$.
- **Column view:** $60 \begin{bmatrix} 10 \\ 11 \end{bmatrix} + 80 \begin{bmatrix} 7 \\ 4 \end{bmatrix}$, price times credits, college by college.

The sizes match for a reason: the credits table's **columns** and the price vector's **entries** are the same two colleges.

Notebook U2-3_Matrices-2_MultiplicationAndPowers, §4, the bills computed

Special Matrices to Know on Sight

Zero matrix 0 : every entry is 0 , in any size

$$\begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}_{2 \times 2} \quad \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}_{2 \times 3} \quad \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}_{3 \times 1} \quad 0\bar{x} = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix} = \bar{0}$$

Every row is all zeros, so the answer is $\bar{0}$. Also $0A = A0 = 0$: it **wipes out** anything, like 0 .

Identity I : 1s on the diagonal, 0 elsewhere; always square

$$\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}_{2 \times 2} \quad \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}_{3 \times 3} \quad I\bar{x} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} x \\ y \end{bmatrix} = \bar{x}$$

Each row of I picks out one entry of $\bar{x}$. Also $IA = AI = A$: it **changes nothing**, like 1 .

Notebook U2-3_Matrices-1_ArithmeticAndSpecial, §4, the special types displayed

Exercise: Matrix Times Vector

Work with these

$$A = \begin{bmatrix} 2 & 3 \\ 1 & 4 \end{bmatrix}, \quad B = \begin{bmatrix} 1 & 0 & 2 \\ -1 & 3 & 1 \end{bmatrix}, \quad C = \begin{bmatrix} 1 & 2 \\ 0 & 1 \\ 3 & -1 \end{bmatrix}, \quad D = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$$
$$\bar{u} = \begin{bmatrix} 2 \\ -1 \end{bmatrix}, \quad \bar{v} = \begin{bmatrix} 1 \\ 0 \\ 2 \end{bmatrix}, \quad \bar{w} = \begin{bmatrix} 3 \\ 1 \end{bmatrix}$$

Try it

- 1 Compute $A\bar{u}$ by the row view, then again by the column view.
- 2 Which of $B\bar{u}$, $B\bar{v}$, $C\bar{u}$, $C\bar{v}$ exist? Compute the ones that do.
- 3 Compute $D\bar{w}$, $I\bar{w}$, and $0\bar{w}$, and say in words what each matrix did.
- 4 Compute $A^T\bar{u}$. Is it the same as $A\bar{u}$?
- 5 Write $2x + 3y = 9$, $x + 4y = 7$ as $A\bar{x} = \bar{b}$, then check that $\bar{w}$ solves it.

Exercise: Solution

Answers

① Rows: $A\bar{u} = \begin{bmatrix} 2(2) + 3(-1) \\ 1(2) + 4(-1) \end{bmatrix} = \begin{bmatrix} 1 \\ -2 \end{bmatrix}$. Columns:

$2 \begin{bmatrix} 2 \\ 1 \end{bmatrix} - 1 \begin{bmatrix} 3 \\ 4 \end{bmatrix} = \begin{bmatrix} 4 - 3 \\ 2 - 4 \end{bmatrix} = \begin{bmatrix} 1 \\ -2 \end{bmatrix}$. They agree, as they must.

② B is 2×3 , so it needs a vector with 3 entries:

$B\bar{v} = \begin{bmatrix} 1(1) + 0(0) + 2(2) \\ -1(1) + 3(0) + 1(2) \end{bmatrix} = \begin{bmatrix} 5 \\ 1 \end{bmatrix}$, and $B\bar{u}$, $(2 \times \mathbf{3})(\mathbf{2} \times 1)$, is **undefined**.

C is 3×2 , so it needs 2 entries: $C\bar{u} = \begin{bmatrix} 1(2) + 2(-1) \\ 0(2) + 1(-1) \\ 3(2) + (-1)(-1) \end{bmatrix} = \begin{bmatrix} 0 \\ -1 \\ 7 \end{bmatrix}$, a

vector in $\mathbb{R}^3$ from one in $\mathbb{R}^2$. $C\bar{v}$, $(3 \times \mathbf{2})(\mathbf{3} \times 1)$, is **undefined**.

Exercise: Solution (continued)

Answers

- ③ $D\bar{w} = \begin{bmatrix} 0(3) + 1(1) \\ 1(3) + 0(1) \end{bmatrix} = \begin{bmatrix} 1 \\ 3 \end{bmatrix}$: D **swapped the two entries**, as it does to any vector, $D \begin{bmatrix} x \\ y \end{bmatrix} = x \begin{bmatrix} 0 \\ 1 \end{bmatrix} + y \begin{bmatrix} 1 \\ 0 \end{bmatrix} = \begin{bmatrix} y \\ x \end{bmatrix}$. $I\bar{w} = \bar{w}$: the identity **left $\bar{w}$ alone**. $0\bar{w} = \bar{0}$: the zero matrix **sent $\bar{w}$ to the origin**.
- ④ $A^\top = \begin{bmatrix} 2 & 1 \\ 3 & 4 \end{bmatrix}$, so $A^\top \bar{u} = \begin{bmatrix} 4 - 1 \\ 6 - 4 \end{bmatrix} = \begin{bmatrix} 3 \\ 2 \end{bmatrix}$. **Different** from $A\bar{u}$: transposing changes which numbers sit in each row, so it changes the product.
- ⑤ $\begin{bmatrix} 2 & 3 \\ 1 & 4 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 9 \\ 7 \end{bmatrix}$: the coefficients are A , the unknowns $\bar{x}$, the right-hand sides $\bar{b}$. Check: $A\bar{w} = \begin{bmatrix} 2(3) + 3(1) \\ 1(3) + 4(1) \end{bmatrix} = \begin{bmatrix} 9 \\ 7 \end{bmatrix} = \bar{b}$, so $x = 3$, $y = 1$ **solves** the system.

Matrix Times Matrix

Matrix Times Matrix: the Definition

Nothing new: matrix times vector, once per column

Write B by its columns, multiply A by each, and stand the answers side by side:

$$AB = A [\bar{b}_1 \quad \bar{b}_2 \quad \cdots \quad \bar{b}_p] = [A\bar{b}_1 \quad A\bar{b}_2 \quad \cdots \quad A\bar{b}_p]$$

Definition

For A ($m \times n$) and B ($n \times p$), entry (i, j) of AB is **row i of A dotted with column j of B** :

$$(AB)_{ij} = \sum_{k=1}^n a_{ik} b_{kj} = a_{i1}b_{1j} + a_{i2}b_{2j} + \cdots + a_{in}b_{nj}$$

Sizes must match, or there is no product

$(m \times n) \underbrace{(n \times p)}_{\text{equal}} = (m \times p)$: a row of A must be **as long as** a column of B .

Matrix Times Matrix: See Where Everything Goes

2×2 times 2×2 , every entry written out

$$\begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix}_{2 \times 2} \begin{bmatrix} b_{11} & b_{12} \\ b_{21} & b_{22} \end{bmatrix}_{2 \times 2} = \begin{bmatrix} a_{11}b_{11} + a_{12}b_{21} & a_{11}b_{12} + a_{12}b_{22} \\ a_{21}b_{11} + a_{22}b_{21} & a_{21}b_{12} + a_{22}b_{22} \end{bmatrix}_{2 \times 2}$$

i picks the row of A , j the column of B ; k walks **across** and **down**.

Check: column 1 of the answer is just A times column 1 of B

$$\begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix} \begin{bmatrix} b_{11} \\ b_{21} \end{bmatrix} = \begin{bmatrix} a_{11}b_{11} + a_{12}b_{21} \\ a_{21}b_{11} + a_{22}b_{21} \end{bmatrix}$$

A matrix-vector product, exactly as before. Column 2 is A times column 2 of B the same way.

Matrix Times Matrix: Worked, and Impossible

Worked: $(2 \times 2)(2 \times 2)$, every entry one dot product

$$\begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix} \begin{bmatrix} 5 & 6 \\ 7 & 8 \end{bmatrix} = \begin{bmatrix} 1 \cdot 5 + 2 \cdot 7 & 1 \cdot 6 + 2 \cdot 8 \\ 3 \cdot 5 + 4 \cdot 7 & 3 \cdot 6 + 4 \cdot 8 \end{bmatrix} = \begin{bmatrix} 19 & 22 \\ 43 & 50 \end{bmatrix}$$

Worked: $(2 \times 3)(3 \times 2)$

$$\begin{bmatrix} 1 & 0 & 2 \\ 3 & 1 & 1 \end{bmatrix} \begin{bmatrix} 1 & 2 \\ 0 & 1 \\ 1 & 0 \end{bmatrix} = \begin{bmatrix} 3 & 2 \\ 4 & 7 \end{bmatrix}$$

Not square, but $3 = 3$: fine.

Impossible: the inner sizes do not match

- $(2 \times \mathbf{2})(\mathbf{3} \times 2)$: rows of A have 2 entries, columns of B have 3.
- $(2 \times \mathbf{3})(\mathbf{2} \times 3)$: a 2×3 matrix cannot even multiply **itself**.

No zero-padding, no partial answer; numpy raises an error.

Notebook U2-3.Matrices-2.MultiplicationAndPowers, §2, the top-left entry checked by hand

Line It Up: Where Each Entry of $AB = C$ Goes

$$A \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix} \begin{array}{c} \begin{bmatrix} 5 & 6 \\ 7 & 8 \end{bmatrix} B \\ \vdots \\ \begin{bmatrix} 19 & 22 \\ 43 & 50 \end{bmatrix} C \end{array}$$

How to use it

- Put B **above** and A to the **left**; $C = AB$ fills the corner.
- Row i of A runs **across**, column j of B runs **down**, and they cross at c_{ij} .
- c_{21} is row 2 dotted with column 1:
 $3 \cdot 5 + 4 \cdot 7 = 43$.

The layout explains the shapes

C gets A 's **rows** and B 's **columns**, and a row of A must be as long as a column of B : $(m \times n)(n \times p) = (m \times p)$. The matrix-vector picture is the case where B has one column.

Rules, and the One That Fails

What carries over from numbers

- **Associative:** $(AB)C = A(BC)$
- **Distributive:** $A(B + C) = AB + AC$
- **Identity:** $IA = AI = A$
- **Zero:** $0A = A0 = 0$
- **Transpose of a product:**
 $(AB)^T = B^T A^T$, the order **reverses**

What does not: $AB \neq BA$

$$A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}, \quad B = \begin{bmatrix} 5 & 6 \\ 7 & 8 \end{bmatrix}$$

$$AB = \begin{bmatrix} 19 & 22 \\ 43 & 50 \end{bmatrix} \neq \begin{bmatrix} 23 & 34 \\ 31 & 46 \end{bmatrix} = BA$$

The shapes can differ too

- $(2 \times 3)(3 \times 2) = 2 \times 2$, but $(3 \times 2)(2 \times 3) = 3 \times 3$.
- $(2 \times 3)(2 \times 3)$: **undefined**, in either order.

The habit to build: never swap the order of a matrix product without a reason.

Notebook U2-3_Matrices-2_MultiplicationAndPowers, §3, AB and BA compared

Why $AB \neq BA$

The top-left entry of each, with $A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$ and $B = \begin{bmatrix} 5 & 6 \\ 7 & 8 \end{bmatrix}$

$$(AB)_{11} = \underbrace{\begin{bmatrix} 1 & 2 \end{bmatrix}}_{\text{row 1 of } A} \underbrace{\begin{bmatrix} 5 \\ 7 \end{bmatrix}}_{\text{column 1 of } B} = 1 \cdot 5 + 2 \cdot 7 = 19$$

$$(BA)_{11} = \underbrace{\begin{bmatrix} 5 & 6 \end{bmatrix}}_{\text{row 1 of } B} \underbrace{\begin{bmatrix} 1 \\ 3 \end{bmatrix}}_{\text{column 1 of } A} = 5 \cdot 1 + 6 \cdot 3 = 23$$

In general

$$(AB)_{ij} = \sum_k a_{ik} b_{kj} \quad \text{vs.} \quad (BA)_{ij} = \sum_k b_{ik} a_{kj}$$

- AB : rows of A with columns of B . BA : rows of B with columns of A .
- Different numbers get paired, so the sums differ. For numbers, ab and ba pair the **same** two.

Geometrically, in Unit 2-4: rotate then flip is not flip then rotate.

Matrix Powers

Definition: a square matrix times itself

$$A^2 = AA, \quad A^3 = AAA, \quad A^n = \underbrace{A A \cdots A}_{n \text{ times}}$$

Only square matrices have powers: $(m \times n)(m \times n)$ needs $n = m$.

Worked: $A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$

$$A^2 = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix} \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix} = \begin{bmatrix} 1+6 & 2+8 \\ 3+12 & 6+16 \end{bmatrix} = \begin{bmatrix} 7 & 10 \\ 15 & 22 \end{bmatrix}$$

Watch out

- A^2 is **not** each entry squared: $\begin{bmatrix} 1 & 4 \\ 9 & 16 \end{bmatrix}$ is wrong. In numpy, `A**2` squares entries; use `A @ A`.
- Powers of **one** matrix commute, $A^2 A^3 = A^3 A^2 = A^5$, but $(AB)^2 \neq A^2 B^2$ in general.

Exercise: Matrix Products

Try it

Let $A = \begin{bmatrix} 1 & 0 \\ 2 & 1 \end{bmatrix}$, $B = \begin{bmatrix} 3 & 1 \\ 0 & 2 \end{bmatrix}$, $C = \begin{bmatrix} 1 & 2 & 0 \\ 0 & 1 & 3 \end{bmatrix}$.

- ① Compute AB and BA . Are they equal?
- ② State the dimensions of AC and compute it.
- ③ Explain why CA does not exist.
- ④ Compute $(AB)^{\top}$ and $B^{\top}A^{\top}$ and confirm they match.
- ⑤ Compute B^2 . Is it the same as squaring each entry of B ?

Exercise: Solution

Answers

$$\textcircled{1} \quad AB = \begin{bmatrix} 1(3) + 0(0) & 1(1) + 0(2) \\ 2(3) + 1(0) & 2(1) + 1(2) \end{bmatrix} = \begin{bmatrix} 3 & 1 \\ 6 & 4 \end{bmatrix},$$
$$BA = \begin{bmatrix} 3 + 2 & 0 + 1 \\ 0 + 4 & 0 + 2 \end{bmatrix} = \begin{bmatrix} 5 & 1 \\ 4 & 2 \end{bmatrix}. \quad \textbf{Not equal.}$$

$$\textcircled{2} \quad A \text{ is } 2 \times 2 \text{ and } C \text{ is } 2 \times 3, \text{ so } AC \text{ is } 2 \times 3: \quad AC = \begin{bmatrix} 1 & 2 & 0 \\ 2 & 5 & 3 \end{bmatrix}. \quad (\text{Row 2: } 2 \begin{bmatrix} 1 & 2 & 0 \end{bmatrix} + 1 \begin{bmatrix} 0 & 1 & 3 \end{bmatrix}.)$$

$\textcircled{3} \quad CA$ would be $(2 \times 3)(2 \times 2)$. The inner dimensions are 3 and 2: a row of C has three entries and a column of A has two, so there is nothing to pair the third entry with. **Undefined.**

Exercise: Solution (continued)

Answers

④ $(AB)^{\top} = \begin{bmatrix} 3 & 6 \\ 1 & 4 \end{bmatrix}$. $B^{\top}A^{\top} = \begin{bmatrix} 3 & 0 \\ 1 & 2 \end{bmatrix} \begin{bmatrix} 1 & 2 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} 3 & 6 \\ 1 & 4 \end{bmatrix}$. They match, and $A^{\top}B^{\top}$ would not: the order reverses under a transpose.

⑤ $B^2 = \begin{bmatrix} 3 & 1 \\ 0 & 2 \end{bmatrix} \begin{bmatrix} 3 & 1 \\ 0 & 2 \end{bmatrix} = \begin{bmatrix} 9+0 & 3+2 \\ 0+0 & 0+4 \end{bmatrix} = \begin{bmatrix} 9 & 5 \\ 0 & 4 \end{bmatrix}$. Squaring each entry gives $\begin{bmatrix} 9 & 1 \\ 0 & 4 \end{bmatrix}$: **different**, because the top-right entry of B^2 mixes a row with a column.

Networks

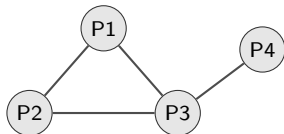
The Adjacency Matrix

Facebook friends: P1 is friends with P2 and P3, P2 with P3, and P3 with P4.

Definition

Draw each person as a node and each friendship as a line. The **adjacency matrix** records it: $a_{ij} = 1$ if i and j are friends, 0 if not.

$$A = \begin{bmatrix} 0 & 1 & 1 & 0 \\ 1 & 0 & 1 & 0 \\ 1 & 1 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{bmatrix}$$



What the matrix shows

- **Symmetric:** a Facebook friendship is mutual. Diagonal 0: nobody is their own friend.
- Each **row sum** is a person's number of friends, their **degree**: 2, 2, 3, 1, so P3 is the hub.

Notebook U2-3_Matrices-2_MultiplicationAndPowers, §5, the network beside its matrix

Squaring the Matrix Counts Two-Step Routes

For an adjacency matrix the product rule gives $(A^2)_{ij} = \sum_k a_{ik} a_{kj}$, and the term $a_{ik}a_{kj}$ is 1 exactly when k is friends with i **and** with j . Summing over k counts the people in between: $(A^2)_{ij}$ **is the number of two-step routes from i to j** , which for $i \neq j$ is their number of **mutual friends**.

$$A^2 = \begin{bmatrix} 2 & 1 & 1 & 1 \\ 1 & 2 & 1 & 1 \\ 1 & 1 & 3 & 0 \\ 1 & 1 & 0 & 1 \end{bmatrix}$$

Check one entry: row 1 of A is $\begin{bmatrix} 0 & 1 & 1 & 0 \end{bmatrix}$ and column 4 is

$\begin{bmatrix} 0 & 0 & 1 & 0 \end{bmatrix}^T$, so

$(A^2)_{14} = 0 + 0 + 1 + 0 = 1$: one mutual friend, P3.

Friend suggestions

Row 1 of A^2 is P1's mutual-friend counts: **one each** with P2 (via P3), with P3 (via P2), and with P4 (via P3). P4 is not P1's friend, yet they share a friend: exactly how "People You May Know" finds P4 for P1.

The diagonal of A^2 is each person's number of friends (out to a friend and straight back), and $(A^2)_{34} = 0$: P3 and P4 are friends but have no mutual friend.

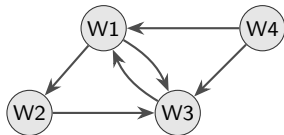
A Directed Network: Website Links

Four web pages: W1 links to W2 and W3, W2 to W3, W3 back to W1, and W4 to W1 and W3.

Definition

A link has a **direction**, so draw it as an arrow. The adjacency matrix records it: $a_{ij} = 1$ if page i **links to** page j , 0 if not. Row = from, column = to.

$$A = \begin{bmatrix} 0 & 1 & 1 & 0 \\ 0 & 0 & 1 & 0 \\ 1 & 0 & 0 & 0 \\ 1 & 0 & 1 & 0 \end{bmatrix}$$



What the matrix shows

- **Not symmetric:** W1 links to W2, but W2 does not link back, so $a_{12} = 1$ while $a_{21} = 0$.
- **Row sums** count links **out**; **column sums** count links **in**. W3 has 3 links in; W4 has none.

Squaring Counts Two-Click Paths

The same product rule, $(A^2)_{ij} = \sum_k a_{ik} a_{kj}$, now follows the arrows: a term is 1 exactly when $i \rightarrow k$ and $k \rightarrow j$.

$$A^2 = \begin{bmatrix} 1 & 0 & 1 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 1 & 1 & 0 \\ 1 & 1 & 1 & 0 \end{bmatrix}$$

Check one entry

$$(A^2)_{41} = \underbrace{\begin{bmatrix} 1 & 0 & 1 & 0 \end{bmatrix}}_{\text{row 4 of } A} \underbrace{\begin{bmatrix} 0 \\ 0 \\ 1 \\ 1 \end{bmatrix}}_{\text{column 1}} = 1$$

One path: W4 $\rightarrow$ W3 $\rightarrow$ W1.

What it says

- $(A^2)_{ij}$ is the number of two-click paths from page i to page j .
- A^2 is not symmetric either: $(A^2)_{41} = 1$ but $(A^2)_{14} = 0$.
- **Column 4 is still all zeros:** no path of any length reaches W4, because nothing links to it.

Higher Powers

The general fact

$(A^n)_{ij}$ counts the **walks of length** n from i to j : sequences of n connections that start at i and end at j . It follows from the same argument applied one step at a time, since $A^n = A \cdot A^{n-1}$.

Two things to keep straight:

- A walk may **revisit** people: $(A^3)_{12}$ includes P1 to P2 to P1 to P2. For routes that never repeat a person, powers overcount.
- The entries **grow fast**. How fast, and which entries dominate, is what **eigenvalues** measure (Unit 2-6), and it underlies Google's PageRank: next slide.

What Else Linear Algebra Does with Networks

With today's tools

- **Triangles:** $\frac{1}{6} \text{tr}(A^3)$ counts friend groups of three (tr sums the diagonal). Our Facebook network has 1.
- **Distance:** the smallest n with $(A^n)_{ij} > 0$. P2 and P4 are 2 steps apart.
- **Random surfing:** divide each row of A by its sum; its powers give the chance of being on each page after n clicks.

With eigenvectors (Unit 2-6)

- **Influence:** $A\bar{x} = \lambda\bar{x}$ scores each person by their friends' scores (**eigenvector centrality**).
- **PageRank:** where random surfing settles in the long run, an eigenvector of the link matrix.
- **Communities:** eigenvectors of the **Laplacian** $L = D - A$ (degrees on D 's diagonal) split a network into tight-knit groups.

And in AI: a **graph neural network** layer multiplies by A , so each node mixes in its neighbors' features.

Review: Matrices

What we covered: the object and its arithmetic

- ① “Is $\bar{b}$ a combination of these vectors?” is $A\bar{x} = \bar{b}$ with the vectors as the **columns** of A . A solution is the list of weights; an impossible equation like $0 = 2$ means $\bar{b}$ is out of reach.
- ② An $m \times n$ **matrix** is a grid with m rows and n columns; the entry a_{ij} is in row i , column j . Rows first, always. A column vector is an $n \times 1$ matrix.
- ③ A **data table** is a matrix: rows are records, columns are features, and both views matter.
- ④ Know on sight the **zero** and **identity** matrices.
- ⑤ **Addition** and **scalar multiplication** are entry by entry and need matching shapes.
- ⑥ The **transpose** swaps rows and columns, $(A^T)_{ij} = a_{ji}$. A is **symmetric** when $A^T = A$, and $A + A^T$ always is.

Review: Matrices

What we covered: multiplication and powers

- ⑦ Two views of $A\bar{x}$: the **column view**, a linear combination of A 's columns weighted by $\bar{x}$, and the **row view**, one **dot product** per entry. A system is now one equation, $A\bar{x} = \bar{b}$.
- ⑧ $I\bar{x} = \bar{x}$ and $0\bar{x} = \bar{0}$: the identity leaves every vector alone, the zero matrix sends every vector to the origin.
- ⑨ $(AB)_{ij}$ is **row i of A dotted with column j of B** ; the shapes must chain as $(m \times n)(n \times p) = m \times p$. In block notation, $AB = [A\bar{b}_1 \ \cdots \ A\bar{b}_p]$: one matrix-vector product per column of B .
- ⑩ “Product” is a name **by analogy**: products are associative and distributive, $(AB)^\top = B^\top A^\top$, but the shapes must fit and $AB \neq BA$ in general.
- ⑪ **Powers** A^n need a square A and are not entry by entry. In a network, $(A^n)_{ij}$ counts the **n -step walks** from i to j .

Notebooks: U2-3_Matrices-1, -2. **Previous:** Unit 2-2. **Next:** Unit 2-4, Transformations.